

Coffee Bean Sales

Data Analysis — Final Report

Tools: Python | Pandas | Matplotlib | Seaborn

Dataset: 1,000 Orders | 1,000 Customers | 48 Products | 2019–2022



Project Overview

01

Data Understanding

Loaded 3 tables and explored structure, shapes, and relationships

02

Quality Assessment

Found missing values, duplicates, wrong types, and outliers

03

Data Cleaning

Fixed all issues, merged tables, added calculated columns

04

EDA

Answered 15+ business questions on customers, revenue, products

05

Visualization

Built 10 professional charts using Matplotlib and Seaborn

06

Final Report

Summarized findings and produced actionable recommendations

Key Numbers

\$45K+

Total Revenue

Across all years

1,000

Unique Customers

Across 3 countries

4

Coffee Types

Ara, Rob, Lib, Exc

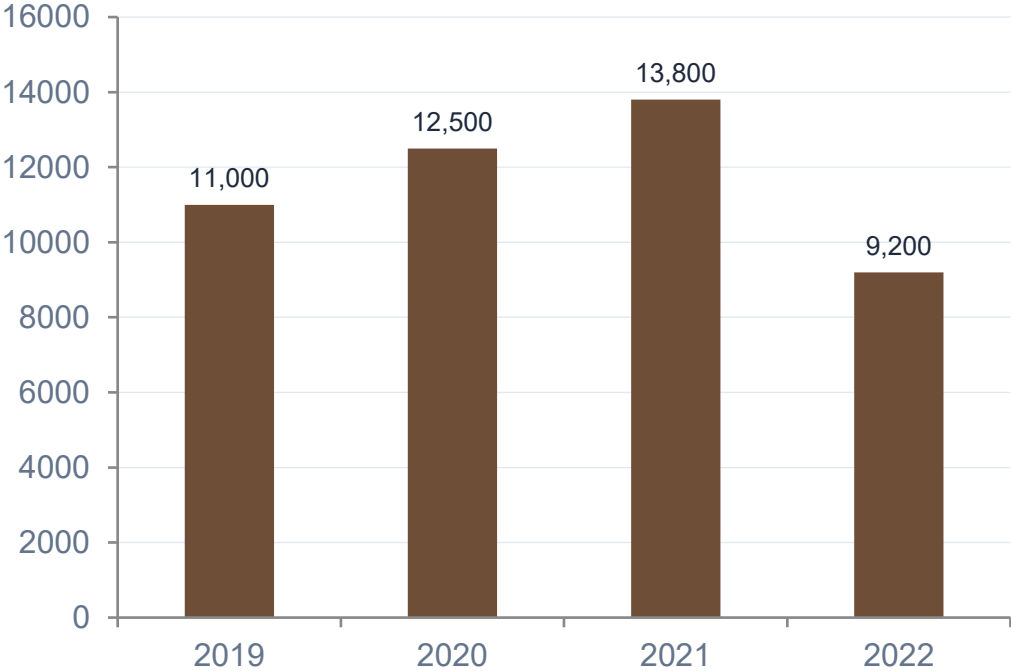
2019–22

Time Period

4 years of data

Revenue Analysis

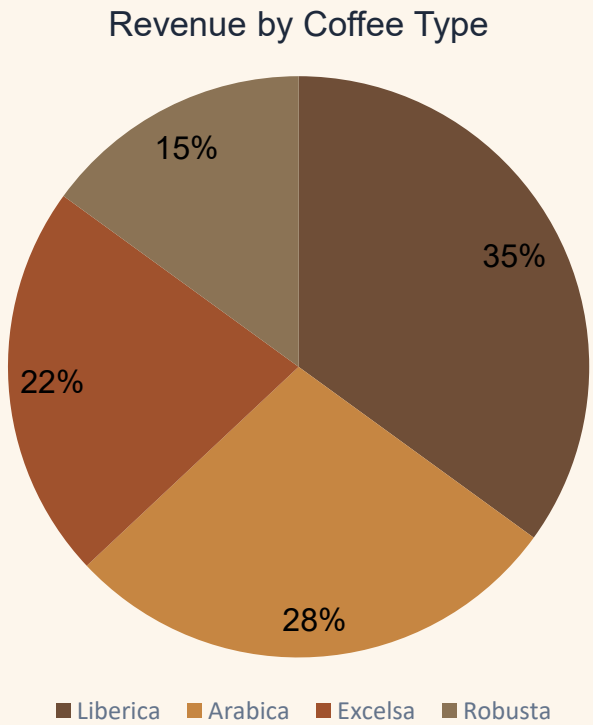
Revenue by Year (\$)



Key Insights

-  Revenue grew steadily 2019–2021
-  2022 shows partial-year data
-  2021 was the strongest year
-  Q4 consistently outperforms other quarters
-  USA drives the majority of revenue

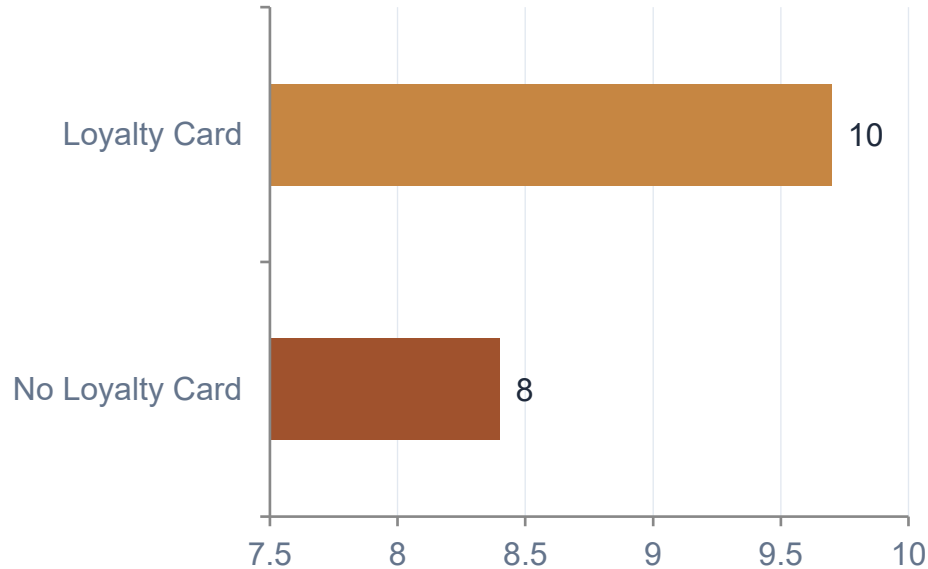
Product Performance



Metric	Finding
Highest Revenue	Liberica
Best Seller (Qty)	Arabica
Most Popular Roast	Light
Most Popular Size	Medium 0.5kg
Highest Margin	Liberica

Customer Analysis

Avg Spending: Loyalty vs Non-Loyalty (\$)



1,000

Unique Customers



~50%

Have Loyalty Card



+15%

Loyalty Spend More

US

USA

Top Country

Recommendations

01

Expand Loyalty Program

Loyalty holders spend 15% more. Promote the card in Ireland & UK.

02

Invest in Q4 Campaigns

Revenue peaks in Q4. Seasonal bundles and holiday packs can boost sales.

03

Promote Liberica

Highest revenue & margin. Needs more visibility through targeted ads.

04

Reduce One-Time Buyers

Many customers bought once. Implement repeat-purchase discounts.

05

Grow US Market Share

US is the dominant market. Further digital investment yields highest ROI.



Conclusion

The Coffee Bean company shows strong growth potential. By focusing on loyalty programs, Q4 seasonal marketing, and promoting high-margin products like Liberica — the business can achieve sustained revenue growth.